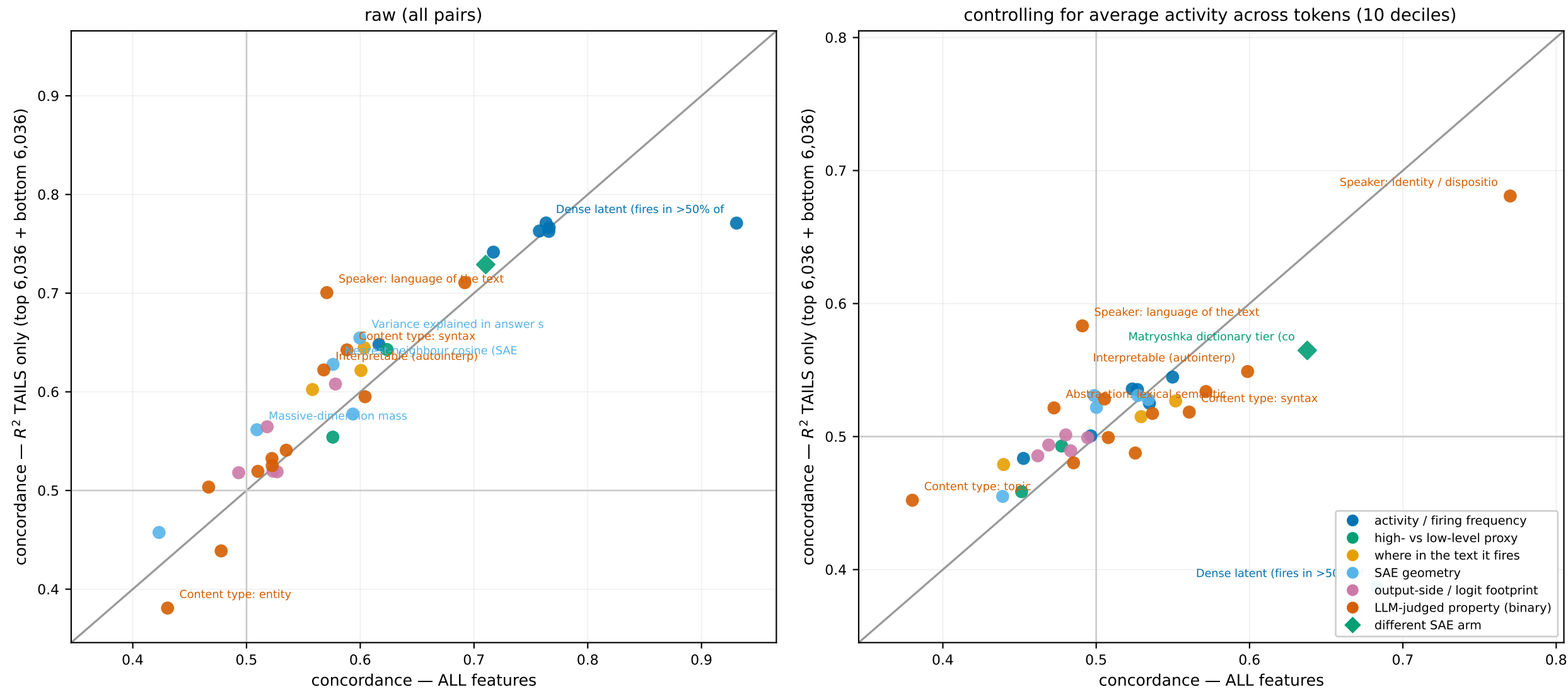


# Are some predictors sharper at the extremes?

tail = the 5.0% best- and 5.0% worst-predicted features by  $R^2$  (12,072 of 120,716) · grey line = no difference



$c = P(\text{the feature with the higher predictor value is the better-predicted one})$ , over pairs the predictor separates; for a binary predictor this is the ordinary AUROC.  
 A point ABOVE the grey line separates the extremes better than it separates the population as a whole; BELOW it, worse. Only the 8 largest movers per panel are labelled.  
 Tail firing-deciles are recomputed WITHIN the tail subpopulation, so the right panel conditions on firing rate among the extremes, not on the whole-population strata.  
 The DIAMOND (matryoshka tier) is a DIFFERENT dictionary (layer-20 SAE lens matryoshka, 16,384-feature panel) — same-direction claim only, not row-to-row comparable.  
 Every value is MARGINAL — the other predictors are not controlled (pairwise correlations reach 0.85). No confidence intervals; the tails cut rare-indicator counts ~10x.